PHARMACOLOGY

Drugs Used in Infertility & Assisted Reproduction

SGL 2
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GUE Block

Pharmacology GUE Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

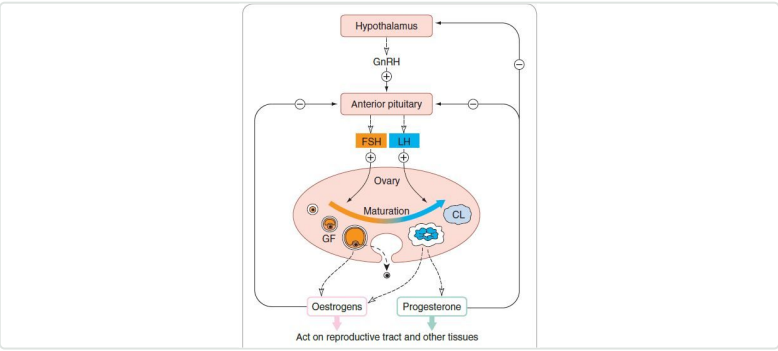
1. How does pulsatile GnRH administration differ in effect from continuous GnRH administration?
2. Which receptor do LH and hCG share, and which hormone signals through a separate receptor of its own?
3. What is the most serious complication of gonadotropin therapy for ovulation induction?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

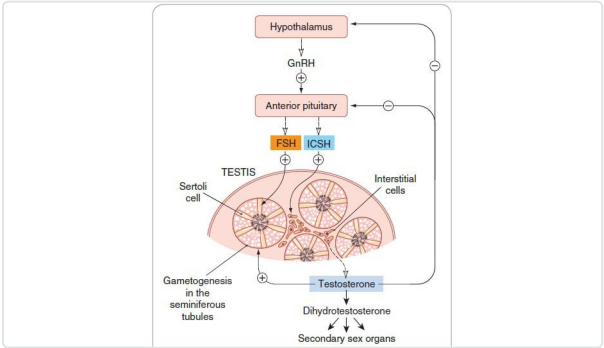
1 Infertility — definitions & causes

- **Infertility** = failure to conceive after 1 year of regular unprotected sex (88–90% of couples conceive within that year).

Female causes	Male causes
Ovulation disorders · tubal abnormality · endometriosis	Azoospermia (impaired spermatogenesis or surgical sterilisation) · oligospermia / impaired sperm motility (e.g. varicocele)



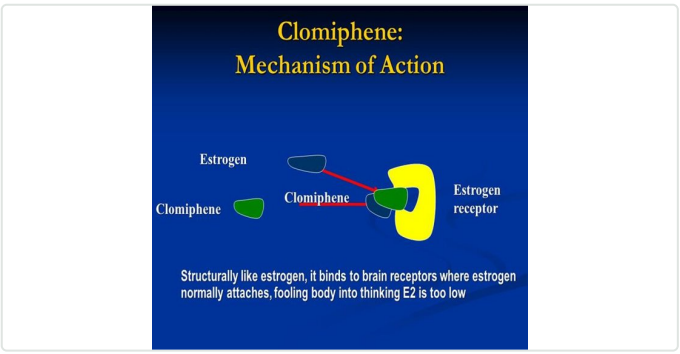
Hormonal control — female reproductive axis (HPG axis)



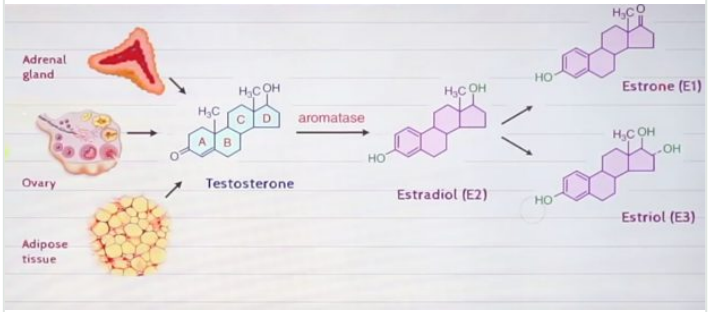
Hormonal control — male reproductive axis

2 Ovulation-inducing drugs — comparison grid

Class	Agent(s)	Mechanism	Clinical use
Anti-oestrogen	Clomiphene citrate (Clomid)	Competitive antagonist — blocks oestrogen receptors in hypothalamus/pituitary → removes negative feedback → ↑FSH, ↑LH → ovulation	PCOS; dysfunctional bleeding with anovulatory cycles; diagnostic (Clomiphene Citrate Challenge Test)
Aromatase inhibitor	Anastrozole, letrozole (Femara)	Non-steroidal reversible competitive inhibitor of aromatase (blocks androstenedione/ testosterone → oestrogen)	1st/2nd line + adjuvant therapy in ER+ postmenopausal breast cancer; infertility due to anovulatory cycle
Gonadotropins	hMG, FSH (uFSH/rFSH), LH (lutropin), hCG / rhCG	Direct replacement of FSH/LH action on the ovary	Anovulatory infertility after clomiphene failure; follicle recruitment for IVF; male hypogonadotrophic hypogonadism
GnRH & analogues	Gonadorelin; agonists (buserelin, goserelin, nafarelin); antagonists (cetrorelix, elagolix)	Pulsatile = stimulates gonadotrophin release; continuous = suppresses it (see discriminator below)	Pulsatile: hypothalamic hypogonadotropic hypogonadism. Continuous: suppress LH surge in IVF protocols; prostate/breast cancer; endometriosis; precocious puberty
Dopamine agonists	Bromocriptine, cabergoline	D2-receptor agonists — inhibit prolactin production	Hyperprolactinemia causing galactorrhea/hypogonadism; pituitary microadenoma



Clomiphene MOA — structurally mimics oestrogen, binds hypothalamic receptors, "fools" the body into reading E2 as low



Biosynthetic pathway: aromatase converts testosterone → oestradiol → oestrone/oestriol — the step aromatase inhibitors block

DISCRIMINATOR — PULSATILE vs CONTINUOUS GnRH (opposite effects, same drug class)

	Effect on gonadotropins	Used for
Pulsatile (SC infusion, mimics physiological secretion)	**Stimulates** FSH/LH release → induces ovulation / spermatogenesis	Hypothalamic hypogonadotropic hypogonadism
Continuous (nasal spray / depot / implant)	**Suppresses** gonadotrophin secretion (after initial flare)	Prostate & breast cancer, endometriosis, precocious puberty; also to block the LH surge before gonadotropin/ hCG in IVF

In prostate cancer, the initial flare in gonadotropin secretion can transiently worsen bone-metastasis pain — start only after an androgen-receptor antagonist (e.g. flutamide).

3 Gonadotropin preparations & the shared-receptor fact

Preparation	Source / form
Menotropins (hMG)	Purified human menopausal gonadotropin — FSH + LH extract
FSH	Urofollitropin (uFSH, from postmenopausal urine) or recombinant (rFSH): follitropin alfa / follitropin beta
LH (lutropin)	Approved only combined with follitropin alfa for profound LH deficiency
hCG	Extracted/purified from urine (IM), or recombinant (rhCG, choriogonadotropin alfa — SC)



Clomiphene citrate (Clomid)



Letrozole (Femara)



hCG injection

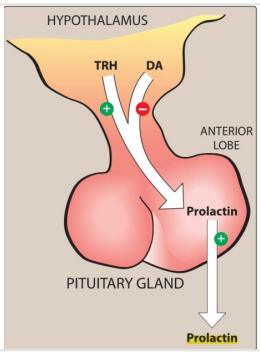
FSH, LH and hCG are all glycoproteins sharing an identical α chain + a distinct β chain that confers receptor specificity. FSH acts through its own **FSH receptor**; **LH and hCG both act through the LH receptor** and are used interchangeably — this is why hCG commonly substitutes for the natural LH surge to trigger ovulation, and why hCG/ rhCG are given to trigger ovulation after follicular recruitment with gonadotropins.

4 Clinical uses & adverse effects of gonadotropins

Clinical uses	Adverse effects (most serious complications)
Anovulatory infertility from hypopituitarism or clomiphene failure; follicle recruitment for IVF (given after endogenous FSH/LH suppression); male infertility from hypogonadotrophic hypogonadism/cryptorchidism; hCG/rhCG trigger ovulation	**Ovarian hyperstimulation syndrome** (ovarian enlargement, ascites, hydrothorax, hypovolemia → shock) · **multiple pregnancy** (15–20% vs ~1% background) · haemoperitoneum (ruptured ovarian cyst) · arterial thromboembolism

5 Hyperprolactinemia & dopamine agonists

- Caused by a pituitary prolactinoma, or by dopamine-antagonist drugs (e.g. metoclopramide, some antipsychotics).



Prolactin secretion: dopamine (DA) inhibits, TRH stimulates



Cabergoline

Drug	Indication	Notes
Bromocriptine	Galactorrhea + hypogonadism; pituitary microadenoma	Oral; ADRs: nausea, vomiting, headache, postural hypotension
Cabergoline	Same indications	**More effective than bromocriptine** for resolving amenorrhea/ oligomenorrhea from galactorrhea, with significantly fewer GI adverse events

P Must-remember pearls

- **Pulsatile GnRH stimulates** gonadotropin release (used for hypothalamic hypogonadotropic hypogonadism); **continuous GnRH suppresses it** (used in prostate/breast cancer, endometriosis, precocious puberty, and to block the LH surge before IVF gonadotropin protocols).
- **FSH** acts through its own receptor; **LH and hCG share the LH receptor** and are used interchangeably — hCG is routinely substituted to trigger ovulation.
- Clomiphene works by structurally mimicking oestrogen and blocking hypothalamic oestrogen receptors, "fooling" the brain into reading oestrogen as low → ↑FSH/LH. Main risks: ovarian enlargement, hot flushes, and multiple gestation (twins).
- **Ovarian hyperstimulation syndrome** is the most serious complication of gonadotropin therapy; multiple pregnancy risk rises to 15–20% (vs ~1% background).
- Cabergoline outperforms bromocriptine for hyperprolactinemia-related amenorrhea/galactorrhea, with fewer GI side effects.

Q Test yourself

1. A woman with PCOS and anovulatory infertility is started on clomiphene citrate. What is its mechanism of action?

A. Direct stimulation of ovarian FSH receptors
B. Inhibition of aromatase
C. Competitive blockade of hypothalamic/pituitary oestrogen receptors, removing negative feedback
D. Dopamine receptor agonism suppressing prolactin
E. Pulsatile GnRH receptor agonism

2. A man with hypothalamic hypogonadotropic hypogonadism is treated with subcutaneous GnRH delivered in pulses to mimic physiological secretion. What effect does this have?

A. It suppresses gonadotropin release
B. It stimulates gonadotropin release and induces spermatogenesis
C. It has no effect unless given continuously
D. It causes permanent pituitary desensitisation
E. It directly stimulates testosterone synthesis without involving LH

3. In an IVF protocol, a GnRH agonist is given continuously before administering gonadotropins. Why?

A. To replace hCG at the end of the cycle
B. To suppress endogenous FSH/LH and prevent a premature LH surge
C. To trigger ovulation
D. To directly stimulate follicular maturation
E. To treat ovarian hyperstimulation syndrome

4. Which hormone acts through its own distinct receptor, separate from the one shared by LH and hCG?

A. Both LH and FSH share one receptor with hCG
B. FSH
C. None — all three share one receptor
D. hCG
E. LH

5. A patient undergoing ovulation induction with gonadotropins develops ovarian enlargement, ascites, and hydrothorax shortly after treatment. Which complication is this?

A. Ovarian hyperstimulation syndrome
B. Ovarian torsion only
C. Mifepristone toxicity
D. GnRH agonist flare
E. Hyperprolactinemia

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — C — Clomiphene is a competitive oestrogen-receptor antagonist at the hypothalamus/pituitary; blocking feedback inhibition raises FSH/LH and triggers ovulation.

2 — B — Pulsatile GnRH mimics physiological secretion and stimulates gonadotropin release, inducing ovulation/spermatogenesis — the opposite of continuous administration, which suppresses it.

3 — B — Pre-treatment with a GnRH agonist (continuous) suppresses endogenous FSH/LH, preventing a premature LH surge before controlled ovarian hyperstimulation with exogenous gonadotropins.

4 — B — FSH acts through the FSH receptor; LH and hCG both act through the shared LH receptor and are used interchangeably.

5 — A — Ovarian enlargement, ascites, hydrothorax and hypovolemia define ovarian hyperstimulation syndrome, the most serious complication of gonadotropin therapy.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam